

# Toward $P \neq NP$ via Time-Bounded Kolmogorov Complexity

A Research Program

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## Abstract

We propose a research program connecting randomized time-bounded Kolmogorov complexity, failures of symmetry of information, chain rules for time-bounded Kolmogorov complexity, and meta-complexity. The program is motivated by three recent developments. Kabanets and Kolokolova (2026) obtained exact complexity-theoretic characterizations of natural chain rules for time-bounded Kolmogorov complexity. Hu, Manor, and Oliveira (2026) proved an unconditional failure of symmetry of information for randomized time-bounded Kolmogorov complexity  $rKt$ , together with a near-optimal lower bound for estimating conditional  $rKt$ . Ilango (2026) established strong random-oracle evidence for the hardness of MCSP and related time-bounded Kolmogorov-complexity problems.

Rather than claiming a proof of  $P \neq NP$ , we isolate three open bridges whose combination would yield such a separation: (i) a one-sided, gap-preserving transfer from randomized  $rKt$  to deterministic  $K^t$ , ideally with a linear gap; (ii) an explicit reduction from SAT to a suitably restricted meta-complexity problem and then to a gap version of conditional  $K^t$ ; and (iii) the circuit-hardness condition  $E \not\subseteq i.o.-SIZE[2^{o(n)}]$ , unless it can be derived from the proposed meta-complexity hardness.

A key correction is that naive direct-product repetition cannot turn a sublinear per-instance gap into a linear gap relative to total input length. Consequently, the most useful first target is a direct one-sided transfer with an  $\Omega(n)$  gap, or a genuinely nontrivial amplification method. The resulting framework is best viewed as a set of independent research problems. Even partial progress on these bridges would be of interest independently of the final  $P$  versus  $NP$  question.

## 1 Introduction

The  $P$  versus  $NP$  problem asks whether every problem whose solutions can be verified in polynomial time can also be solved in polynomial time. Since SAT is NP-complete, it suffices to prove

$$SAT \notin P.$$

Kolmogorov complexity offers a natural language for studying computational information. In the time-bounded setting,  $K^t(x \mid y)$  denotes the length of the shortest program that, given  $y$ , outputs  $x$  within  $t$  computation steps.

Recent work makes this perspective substantially more precise. Kabanets and Kolokolova [1] established exact complexity-theoretic characterizations for natural chain rules for  $K^t$ . Hu, Manor, and Oliveira [2] proved an unconditional failure of symmetry of information for randomized time-bounded Kolmogorov complexity  $rKt$ . Ilango [3] obtained a random-oracle result giving strong evidence for NP-hardness of MCSP and an extension to  $K^t$ .

The purpose of this paper is not to claim a solution, but to formulate a concrete route and identify the precise missing results.

## 2 The Program at a Glance

The proposed route can be summarized as follows.

| Bridge | Target                                                           | Status         | Consequence                                                   |
|--------|------------------------------------------------------------------|----------------|---------------------------------------------------------------|
| I      | $rKt \rightarrow K^t$ with a linear gap                          | Open           | Deterministic chain-rule violation                            |
| II     | $SAT \rightarrow MCSP_{\mathcal{F}} \rightarrow \text{GapMcKtP}$ | Open           | NP-hardness of the target problem                             |
| III    | $E \not\subseteq i.o.-SIZE[2^{o(n)}]$                            | Open / assumed | Eliminates the alternative in the chain-rule characterization |

The full  $P \neq NP$  consequence is conditional on all three bridges. No bridge is claimed here as a theorem.

## 3 Chain Rules and Their Complexity-Theoretic Characterization

For strings  $x_1, \dots, x_\ell, y$ , let

$$N = \sum_{i=1}^{\ell} |x_i|.$$

A natural chain rule for time-bounded Kolmogorov complexity has the form

$$K^t(x_1, \dots, x_\ell \mid y) \geq \sum_{i=1}^{\ell} K^{t^c}(x_i \mid y, x_1, \dots, x_{i-1}) - \ell O(\log t) - e(N, t).$$

The regime of interest is  $e(N, t) = o(N)$ .

Kabanets and Kolokolova [1] show that, for the relevant formulation, this chain rule is equivalent to the conjunction of two conditions:

$$(a) \quad E \not\subseteq i.o.-SIZE[2^{o(n)}],$$

and

$$(b) \quad \text{GapMcKtP} \in \text{promise-P}.$$

Here GapMcKtP denotes the corresponding promise problem for distinguishing sufficiently low conditional time-bounded Kolmogorov complexity from values separated by the relevant sublinear gap.

Under an additional NP-hardness assumption for the gap problem, condition (b) becomes equivalent to  $P = NP$ . Thus the chain rule is tied to a conjunction, rather than to  $P = NP$  alone.

This logical structure is essential. A violation of the chain rule establishes only

$$\neg(a) \vee \neg(b),$$

not  $\neg(b)$  alone. Therefore, a chain-rule violation cannot by itself be converted into  $P \neq NP$  without separately ruling out the failure of (a).

## 4 Randomized Time-Bounded Complexity as the Starting Point

Let  $rKt$  denote randomized time-bounded Kolmogorov complexity. For ordinary Kolmogorov complexity, symmetry of information gives, up to logarithmic terms,

$$K(x, y) = K(x) + K(y \mid x).$$

The corresponding identity is substantially more delicate under time bounds.

Ronneburger [5] established that symmetry of information fails for  $Kt$ . The gap in that result is not, by itself, sufficient to settle  $P$  versus  $NP$ .

Hu, Manor, and Oliveira [2] proved an unconditional failure of symmetry of information for  $rKt$ , and also obtained a near-optimal lower bound for estimating conditional  $rKt$ . Their result supplies an unconditional computational-information obstruction, but it does not directly give a corresponding linear-gap statement for deterministic  $K^t$ .

The central question of this program is therefore whether the randomized obstruction can be transferred to deterministic  $K^t$  with a gap large enough to interact with the chain-rule characterization.

## 5 Target I: One-Sided Local Transfer from $rKt$ to $K^t$

A full equivalence between  $rKt$  and  $K^t$  is unnecessarily strong. The proposed target is a local transfer for an explicitly constructed family of instances.

**Target 1** (One-Sided Gap-Preserving Transfer). *Find a polynomial-time transformation*

$$T : (x, y, s) \mapsto (x', y', s')$$

*and a polynomial time bound  $q$  such that, for a designated efficiently constructible family,*

$$rKt(x \mid y) > s + \Delta(n)$$

*implies*

$$K^{q(t)}(x' \mid y') > s' + \Omega(\Delta(n)).$$

*The preferred regime is  $\Delta(n) = \Omega(n)$ .*

The lower-bound direction is the preferred first target. The corresponding constructive direction,

$$rKt(x \mid y) \leq s \implies K^{q(t)}(x' \mid y') \leq s' + O(\log n),$$

is considerably stronger. It requires turning an efficiently describable randomized object into a comparably short deterministic description, closely related to the derandomization and pseudodeterministic difficulties surrounding these constructions.

For that reason, the program does not require the constructive direction at the first stage.

## 6 The Scaling Obstruction to Naive Amplification

Suppose a single instance has gap  $\Delta(n)$ , and suppose  $m$  independent copies satisfy a direct-product inequality of the form

$$\text{Gap}(X, Y) \geq m\Delta(n) - O(m \log n),$$

where the total input length is  $N = mn$ .

To violate a chain rule with an  $o(N)$  error term, the final gap must be  $\Omega(N)$ . But

$$\frac{m\Delta(n)}{mn} = \frac{\Delta(n)}{n}.$$

Therefore, if  $\Delta(n) = o(n)$ , the relative gap remains  $o(1)$  no matter how many independent copies are concatenated.

Consequently, direct-product repetition does not transform a sublinear per-instance gap into a linear gap relative to total input length.

The meaningful alternatives are:

1. prove the transfer directly with  $\Delta(n) = \Omega(n)$ ; or
2. construct a genuinely nontrivial combiner that increases the gap-to-length ratio.

## Revised Target I'

Prove the one-sided transfer above with a linear gap, or develop a nontrivial amplification mechanism that converts a sublinear local gap into a linear global gap. Either result would constitute independent progress in time-bounded Kolmogorov complexity.

## 7 Target II: Restricted Meta-Complexity Reduction

The second bridge concerns the reduction from SAT to gap conditional  $K^t$ .

MCSP asks whether a Boolean function, represented by its truth table, has a circuit of size at most a given threshold. The NP-completeness of standard MCSP remains open. Huang, Ilango, and Ren [4] proved NP-hardness of approximating meta-complexity under cryptographic assumptions, including essentially optimal hardness for conditional time-bounded Kolmogorov complexity in an appropriate regime. Ilango [3] subsequently gave a random-oracle result with a P/poly reduction from an NP-hard approximation problem to MCSP on circuits with a uniformly random oracle, with a corresponding extension to  $K^t$ .

Instead of attempting to derandomize unrestricted MCSP immediately, we propose restricting attention to a structured class  $\mathcal{F}$ .

**Target 2** (Restricted Meta-Complexity Reduction). *Construct an explicit class  $\mathcal{F}$  such that*

$$SAT \leq_p MCSP_{\mathcal{F}} \leq_p \text{GapMcKtP},$$

*while maintaining useful structural properties such as efficient samplability, an efficiently testable representation, and sufficient structure for the reduction.*

The restriction is not intended to trivialize the problem. It is intended to isolate the portion of meta-complexity hardness actually needed by the overall route.

## 8 Target III: Derandomization of the Relevant Gap Property

Ilango's result is a random-oracle statement. The proposed program therefore does not require reproducing the entire random oracle. Instead, the derandomization target should preserve the specific gap property used by the reduction.

Let  $G^{\mathcal{O}}(f)$  denote the circuit-gap quantity employed by the random-oracle reduction, and let  $G^h(f)$  denote the value after replacing the oracle  $\mathcal{O}$  by an explicit family  $h$ . The desired statement is of the form

$$G^{\mathcal{O}}(f) \geq \Delta \implies G^h(f) \geq \Delta - O(\log n),$$

for the instances produced by the reduction.

The preservation requirement is deliberately narrow: the replacement family need not simulate the full distribution of the random oracle, reproduce every property of the oracle, or yield a complete deterministic analogue of Ilango's theorem. It only needs to preserve the circuit-size separation that the reduction actually uses to distinguish its two cases, on the relevant efficiently generated instances.

A successful construction would therefore establish the part of derandomization needed by Target II without solving the much broader problem of constructing a deterministic surrogate for a random oracle.

## 9 The Remaining Logical Gap

Assume that Revised Target I' holds. Because symmetry of information fails unconditionally for  $rKt$ , the transfer would then yield a family exhibiting a linear failure of the corresponding deterministic  $K^t$  chain rule:

$$K^t(X, Y) < K^{t^c}(X) + K^{t^c}(Y \mid X) - \Omega(N),$$

in an appropriate formulation.

By the Kabanets–Kolokolova characterization, this contradicts the chain rule and therefore implies

$$\neg(a) \vee \neg(b),$$

where

$$(a) : E \not\subseteq i.o.-SIZE[2^{o(n)}]$$

and

$$(b) : \text{GapMcKtP} \in \text{promise-P}.$$

To conclude that (b) fails, we need the auxiliary hypothesis

$$(H) \quad E \not\subseteq i.o.-SIZE[2^{o(n)}].$$

This is a standard hardness assumption in the surrounding literature, but it is not established by the present program and should be tracked explicitly unless it can be derived from Target II or another independent result.

Under (H), the chain-rule violation implies

$$\text{GapMcKtP} \notin \text{promise-P}.$$

If Target II establishes

$$SAT \leq_p \text{GapMcKtP},$$

then  $\text{GapMcKtP} \in P$  would imply  $SAT \in P$ , and hence  $P = NP$ . Therefore the failure of polynomial-time solvability yields

$$P \neq NP.$$

The complete conditional route is therefore

$$rKt \xrightarrow{\text{linear-gap transfer}} K^t, \quad SAT \longrightarrow MCSP_{\mathcal{F}} \longrightarrow \text{GapMcKtP},$$

together with the auxiliary hardness condition (H).

## 10 Three Open Bridges

The program can now be stated compactly.

**Bridge I – Randomized to deterministic information.** Prove a one-sided, local, linear-gap transfer  $rKt \rightarrow K^t$ .

**Bridge II – Structured meta-complexity hardness.** Prove  $SAT \leq_p MCSP_{\mathcal{F}} \leq_p \text{GapMcKtP}$  for an explicit, sufficiently structured family  $\mathcal{F}$ .

**Bridge III – Eliminate the alternative in the chain-rule characterization.** Establish  $E \not\subseteq i.o.-SIZE[2^{o(n)}]$ , or prove that the required meta-complexity hardness already implies it.

If all three bridges are established, the resulting chain yields  $SAT \notin P$ , and therefore  $P \neq NP$ . No such conclusion is asserted unconditionally here.

## 11 Weaker Intermediate Problems

The full program may be too ambitious to attack directly. Several weaker problems are nevertheless meaningful:

1. establish the one-sided transfer with any nontrivial gap, and separately determine whether a genuine combiner can amplify it; a direct product alone is insufficient;
2. investigate restricted computational models, such as bounded-depth circuits or branching programs;
3. consider average-case versions over efficiently samplable distributions;
4. allow polynomial advice in the transfer transformation; and
5. determine whether NP-hardness of the relevant gap meta-complexity problem already implies (H).

Partial progress on any of these questions would clarify where the randomized-to-deterministic conversion becomes genuinely difficult.

## 12 Relation to Known Barriers

The program does not attempt to prove a direct lower bound against arbitrary polynomial-size circuits. Instead, it routes the problem through computational information and meta-complexity.

This is potentially useful because meta-complexity problems encode properties of computational descriptions rather than only combinatorial properties of Boolean functions. At the same time, this is not an escape from known barriers: MCSP and related meta-complexity problems have themselves resisted longstanding hardness questions.

The proposed contribution is therefore architectural rather than definitive. It identifies a precise interface between randomized computational information, time-bounded Kolmogorov complexity, and meta-complexity.

## 13 Conclusion

This paper does not claim a proof of  $P \neq NP$ . Instead, it formulates a concrete research program based on recent unconditional and complexity-theoretic results.

The central insight is that the unconditional failure of symmetry of information for randomized  $rKt$  provides a potentially useful source of hardness, but converting it into a deterministic  $K^t$  obstruction requires a substantially stronger transfer than currently known.

The scaling analysis also rules out a simple amplification strategy: independent direct products preserve the gap-to-length ratio. Consequently, the decisive transfer target is naturally a linear-gap one-sided theorem, unless a new nontrivial combiner can be found.

The second major challenge is meta-complexity hardness. A restricted reduction from SAT through a structured MCSP variant could isolate the exact hardness needed without requiring an immediate solution of unrestricted MCSP. The third challenge is logical rather than merely technical: the chain-rule characterization contains an independent circuit-hardness condition, which must either be assumed or derived.

Thus the route to  $P \neq NP$  is best represented not as a single conjectural proof but as three explicit bridges. Their combination would establish  $SAT \notin P$ , and hence  $P \neq NP$ . Each bridge is independently open, and progress on any one of them would be meaningful even without resolving the Millennium Problem.

## References

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